

Name: _____

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Course: Deep Learning (MMC321)

Test: Partial Answers

Q1. Define the core principles of Deep Learning and explain key architectural layers. [2 Marks]

Deep Learning uses neural networks with many layers to learn patterns from data. It has input layer, hidden layers and output layer. Hidden layers use weights and activation functions to process the input.

Q2. Describe the fundamental workflow and mechanisms in Deep Learning. [2 Marks]

First data is collected and given to the neural network. Then forward propagation produces output and loss is calculated. Back propagation changes the weights to improve the result.

Q3. Explain key algorithms or modeling techniques used in Deep Learning with examples. [2 Marks]

CNN is used for images and RNN can be used for sequence data. Back propagation is used to train the network.

Q4. Analyze the transaction/query optimization and concurrency control in Deep Learning. [2 Marks]

Concurrency control is used when many users access a system at the same time. Query optimization makes database queries faster.

Q5. Compare different design approaches in Deep Learning with their pros and cons. [2 Marks]

CNN works well with images. RNN is useful for sequences. CNN can learn spatial features, while RNN can remember previous information.

Q6. Solve a practical implementation scenario using Deep Learning best practices. [5 Marks]

For image classification, I would collect images and divide them into training and test data. Then I would train a CNN using an optimizer. After training I would check its accuracy and use the model for prediction.

Q7. Discuss security, integrity constraints, and error recovery in Deep Learning. [5 Marks]

Security means protecting the model and data from unauthorized users. If an error happens, the system should save the model and restore it. Integrity is important so that the data is not changed incorrectly.

Q8. Design a scalable solution architecture for a real-world Deep Learning system. [5 Marks]

A system can have data storage, preprocessing, model training and an API. GPU can be used for training. More servers can be added when there are many users. The model should also be monitored.

Q9. Define the core principles of Deep Learning and explain key architectural layers. [5 Marks]

Deep Learning is a part of machine learning which uses neural networks.

The input layer receives data and hidden layers process it. The output layer gives the prediction. ReLU is one activation function.

Q10. Describe the fundamental workflow and mechanisms in Deep Learning. [10 Marks]

Deep Learning starts with collecting data and preprocessing it. The data is divided into training and testing sets.

During forward propagation the input goes through the layers and produces an output. Loss is calculated and back propagation is used to calculate gradients.

The optimizer updates the weights and training is repeated for many epochs. After training the model is tested on new data.

Q11. Explain key algorithms or modeling techniques used in Deep Learning with examples. [10 Marks]

CNN is commonly used for image classification because convolution can learn image features. RNN can process sequence data.

LSTM is a type of RNN which uses gates and can remember information for longer time. Transformers use attention and are useful for language tasks.

Back propagation is used for learning the weights of neural networks.